

EN3150 Assignment 02: Learning from data and related challenges and classification

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Instructions. Answer all questions. Show mathematical steps, provide clearly commented Python code, label all plots, and use a fixed random seed whenever randomness is involved. Unless otherwise stated, use an 80:20 train-test split.

1 Linear Regression, Outliers, and Regularization [30 marks]

Use Listing 1 to generate the synthetic regression dataset. Here, use your index no (without letter) as an input.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from sklearn.model_selection import train_test_split
4 from sklearn.linear_model import LinearRegression, Ridge, Lasso
5 from sklearn.metrics import mean_squared_error
6 from sklearn.preprocessing import StandardScaler
7 from sklearn.pipeline import Pipeline
8
9 rng = np.random.default_rng(indexno)
10 X = rng.uniform(-3, 3, size=(160, 1))
11 y = 2.5 + 1.8 * X[:, 0] + rng.normal(0, 0.8, size=160)
12
13 # Add large errors to 12 observations.
14 outlier_idx = rng.choice(len(y), size=12, replace=False)
15 y[outlier_idx] += rng.uniform(10, 15, size=12)
```

Listing 1: Synthetic regression data generation with deliberate outliers.

1. Explain why Ordinary Least Squares (OLS) estimation is particularly sensitive to severe vertical outliers in y . Reference the squared loss function $J(w) = \frac{1}{2N} \sum_{i=1}^N (y_i - w^T x_i)^2$ in your answer. [4]
2. Fit an ordinary least-squares (OLS) model to the generated data. Report its fitted intercept (w_0), slope coefficient (w_1), training Mean Squared Error ($\text{MSE}_{\text{train}}$), and test Mean Squared Error (MSE_{test}). Provide a scatter plot showing the data points along with the fitted regression line. [5]
3. Define $a_i \in (0, 1]$ as the sample-specific weight, where $a_i = 0.05$ for the known outliers and $a_i = 1.0$ for all inliers.

Notation

- N : Total number of training instances
- D : Number of feature dimensions

- $\mathbf{X} \in \mathbb{R}^{N \times (D+1)}$: Data/Design matrix with an added lead column of ones for the intercept bias.
- $\mathbf{y} \in \mathbb{R}^N$: Target response vector $\mathbf{y} = [y_1, y_2, \dots, y_N]^T$.
- $\mathbf{w} \in \mathbb{R}^{D+1}$: Parameter weight vector $\mathbf{w} = [w_0, w_1, \dots, w_{D+1}]^T$.
- $\mathbf{A} \in \mathbb{R}^{N \times N}$: Positive-definite diagonal weighting matrix $\mathbf{A} = \text{diag}(a_1, a_2, \dots, a_N)$.

Derive the closed-form matrix expression for the weighted least-squares estimator:

$$\hat{\mathbf{w}}_{\text{WLS}} = (\mathbf{X}^T \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{A} \mathbf{y}.$$

Implement this WLS estimator using NumPy and compare its fitted line against standard OLS. [7]

4. Regularization in High Dimensions:

Create an expanded dataset containing nine pure noise variables:

```

1 X_noise = rng.normal(size=(X.shape[0], 9))
2 X10 = np.column_stack([X, X_noise])
3

```

Fit OLS, Ridge Regression (L_2), and Lasso Regression (L_1). Tune the regularization strength hyperparameter $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ using 5-fold cross-validation. Report optimal α^* values and cross-validated test MSEs. [8]

5. Compare Ridge and Lasso in terms of their mathematical penalty terms ($\|\mathbf{w}\|_2^2$ vs. $\|\mathbf{w}\|_1$), resulting coefficient magnitudes, and capacity for sparse feature selection. Explain which model is preferable when $D \gg N$ and only a small subset of features are genuinely informative. [4]
6. Explain analytically why standardized feature scaling ($z = \frac{x - \mu}{\sigma}$) is mandatory prior to penalizing weights in Ridge or Lasso regression. [2]

2 Logistic Regression: Preprocessing, Interpretation, and Evaluation [35 marks]

Use Listing 2 for data generation.

```
1 import numpy as np
2 import pandas as pd
3 from sklearn.datasets import make_classification
4
5 X_num, y = make_classification(
6     n_samples=600,
7     n_features=4,
8     n_informative=3,
9     n_redundant=1,
10    class_sep=0.9,
11    weights=[0.68, 0.32],
12    flip_y=0.03,
13    random_state=21
14 )
15
16 df = pd.DataFrame(X_num, columns=["x1", "x2", "x3", "x4"])
17 rng = np.random.default_rng(21)
18 df["region"] = rng.choice(["north", "south", "east"], size=len(df))
19 df["device"] = rng.choice(["mobile", "desktop"], size=len(df))
20 df.loc[rng.choice(len(df), 25, replace=False), "x3"] = np.nan
```

Listing 2: Classification data containing mixed attributes and missing values.

1. Explain why passing string-valued columns directly to LogisticRegression throws a runtime error. Furthermore, explain why applying basic LabelEncoder mapping categories to integers $\{0, 1, 2\}$ introduces a spurious ordinal assumption that degrades model generalizability. [5]
2. Construct a scikit-learn Pipeline combined with ColumnTransformer that:
 - Imputes missing numerical entries with training set column medians;
 - Standardizes numerical features ($\mu = 0, \sigma = 1$);
 - Imputes missing categorical features using the most frequent mode;
 - One-hot encodes categorical attributes; and
 - Fits a Logistic Regression classifier.

Split the dataset into training (80%) and test (20%) subsets using a stratified split with `random_state=42`. [8]
3. Perform a grid search over regularization strengths $C \in \{0.01, 0.1, 1, 10, 100\}$ and penalty types $\text{penalty} \in \{\ell_1, \ell_2\}$ using `solver='liblinear'` and 5-fold Stratified Cross-Validation. Optimize for F_1 -score. Report best parameters and cross-validation score. [6]
4. On the test set, evaluate and report accuracy, precision, recall, F_1 -score, and confusion matrix. Explain why overall accuracy can be deeply misleading when dealing with class-imbalanced datasets (e.g., 68% negative, 32% positive). [7]
5. Adjust the classification decision boundary threshold from $p_{\text{threshold}} = 0.50$ down to 0.35. Recompute precision and recall. Describe the Precision-Recall trade-off and outline a realistic domain application (e.g., medical diagnostics, fraud detection) where lowering the threshold is preferred. [4]

6. A logistic regression model predicting whether a student passes with an A^+ ($y = 1$) yields:

$$\ln\left(\frac{p}{1-p}\right) = -7.2 + 0.08x_1 + 1.4x_2,$$

where x_1 denotes study hours, x_2 denotes GPA, and $p = P(y = 1|x)$.

(a) **Direct Probability Computation:**

Derive the sigmoid expression $p = \frac{1}{1+e^{-z}}$ from logit $z = \ln\left(\frac{p}{1-p}\right)$. Calculate the probability p for a student with $x_1 = 45$ study hours and $x_2 = 3.4$ GPA.

- (b) For a target probability $p = 0.70$ and GPA $x_2 = 3.4$, calculate the required study hours x_1 .

3 Optimization of Logistic Regression

[35 marks]

Use Listing 3 to generate the binary classification data. Do not use high-level estimators like `sklearn.linear_model.LogisticRegression` for questions 1–6 in this section.

```
1 import numpy as np
2 from sklearn.datasets import make_blobs
3 from sklearn.model_selection import train_test_split
4 from sklearn.preprocessing import StandardScaler
5
6 X, y = make_blobs(
7     n_samples=1200,
8     centers=[[-2.2, -0.5], [2.0, 1.2]],
9     cluster_std=[1.45, 1.65],
10    random_state=8
11 )
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, stratify=y, random_state=42
14 )
```

Listing 3: Binary dataset for gradient optimization algorithms.

Let $\mathbf{X} \in \mathbb{R}^{N \times (D+1)}$ be the augmented feature matrix including a bias column of ones, $\mathbf{y} \in \{0, 1\}^N$ be target labels, $\mathbf{w} \in \mathbb{R}^{D+1}$ be model parameters, and $p_i = \sigma(\mathbf{w}^T \mathbf{x}_i) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}_i}}$.

1. Derive the gradient vector $\nabla J(\mathbf{w})$ and Hessian matrix $\mathbf{H}$ for Binary Cross-Entropy loss:

$$J(\mathbf{w}) = -\frac{1}{N} \sum_{i=1}^N [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)].$$

2. Write Python functions for numerically stable sigmoid activation and cross-entropy loss computation. Explain why raw implementations of $\ln(p)$ fail when $p \rightarrow 0$ or $p \rightarrow 1$, and why techniques like clipping probabilities ($\epsilon = 10^{-15}$) or utilizing `np.logaddexp` prevent underflow/overflow. [4]
3. Initialize weights to zero $\mathbf{w}_{(0)} = \mathbf{0}$. Implement Batch Gradient Descent using parameter updates:

$$\mathbf{w}_{(t+1)} \leftarrow \mathbf{w}_{(t)} - \eta \nabla J(\mathbf{w}_{(t)}),$$

with learning rate $\eta = 0.1$ and maximum iterations $T = 300$. Include early stopping when $|J_t - J_{t-1}| < 10^{-8}$. Report final weights, total iterations required, and test set accuracy. Here, J_t : The computed loss value at the current iteration t , evaluated as $J(\mathbf{w}_{(t)})$. [6]

4. Implement Mini-Batch Gradient Descent with batch size $B = 32$, learning rate $\eta = 0.05$, and maximum epochs $E = 100$. Shuffle the training set indices at the beginning of every epoch. Plot training loss progression alongside Batch Gradient Descent. [5]

5. Implement damped Newton's optimization method using the update step:

$$\mathbf{w}_{(t+1)} \leftarrow \mathbf{w}_{(t)} - (\mathbf{H} + 10^{-6} \mathbf{I})^{-1} \nabla J(\mathbf{w}_{(t)}),$$

where $10^{-6} \mathbf{I}$ guarantees numerical invertibility. Train for at most 20 iterations using the same early stopping condition. Use `np.linalg.solve` to directly compute search directions rather than computing explicit matrix inverses. [6]

6. Plot loss versus iteration for both Batch Gradient Descent and Newton's Method on a single logarithmic-scale plot. Compare convergence velocity (number of iterations), computational complexity per step ($O(ND)$ vs. $O(ND^2 + D^3)$), and numerical stability. [4]

7. Multiply the second feature of $\mathbf{X}$ by 100 ($x_2 \leftarrow 100 \cdot x_2$) without standardizing features. Run Batch Gradient Descent on this unscaled dataset. Explain why the loss oscillates or fails to converge, referencing the condition number $\kappa(\mathbf{H}) = \frac{\lambda_{\max}}{\lambda_{\min}}$ of the loss surface Hessian. Propose two concrete remedies. [3]

4 Submission

- Upload a report as a pdf file named as "your_indexno_EN3150_A02.pdf". Include the index number and the name within the report as well. Please include all your answers in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism¹ or cheating will not be tolerated.
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¹<https://en.wikipedia.org/wiki/Plagiarism>